BILHA-CATHERINE GITHINJI

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Research Experience

Massachusetts Institute of Technology: CSAIL — Cambridge, MA

Interactive Robotics Group, Graduate Research Assistant — Sep 2019 – Present

HINTs: Human-INTuited Cues for Reinforcement Learning, G. & Shah 2025
Adaptive State Representations for Multi-step Deformable Object Manipulation, G. & Shah 2023
Model-based Control for Robot Manipulation Tasks with High-dimensional State Spaces (SM Thesis 2022)
Generalising Deformable Object Manipulation by Learning to Learn from Demos, G. & Shah 2021
A Roadmap for Climate-Relevant Robotics Research, Alan Papalia et al. 2025

Apple Corporation: Machine Learning Research Group — Cupertino, CA

Neural Research Group, Research Intern — Jun 2022 – Oct 2022

University of Washington: Center for Sensorimotor and Neural Engineering — Seattle, WA

Daniel Lab, Undergraduate Research Assistant — Feb 2013 – Jun 2016

Industry Experience

Microsoft: Vision Group — Redmond, WA

HoloLens, Software Engineer II — Jul 2016 – May 2019

NASA: Software, Robotics, and Simulation Division — Houston, TX

Robotic Systems Technology Branch (ER4), Student Trainee — Jun 2014 – Sep 2015

Education

Massachusetts Institute of Technology — Cambridge, MA

Doctor of Philosophy, Electrical Engineering and Computer Science — Sep 2022 – Present

Specialisation in *Artificial Intelligence*

Accenture Fellow (2024), Alfred Sloan UCEM scholar (2020-25)

Massachusetts Institute of Technology — Cambridge, MA

Master of Science (SM), Electrical Engineering and Computer Science — Sep 2019 – May 1022

Specialisation in *Artificial Intelligence* — GPA: 5.0/5.0

MIT Lemelson Presidential Fellow (2019)

University of Washington — Seattle, WA

Bachelor of Science, Electrical Engineering — Mar 2016

Specialisation in *Digital Signal and Image Processing* with a minor in Mathematics

Interests

Robotics:	dexterous manipulation, force-based control, robust control, assembly
Reinforcement Learning:	model-based policy learning, self-supervision, sample-based optimisation
Machine Learning:	multi-task learning, representation learning, statistical learning theory
Software:	Pytorch, MuJoCo, OpenAi Gym, IsaacLab, ROS, Python, C/C++, Linux
Leisure:	electric/acoustic guitar, running, billiards, language (swahili, french)